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ectopic pregnancy

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Ectopic pregnancy

is defined as implantation of a conceptus outside the normal uterine cavity.

The incidence of ectopic pregnancy in the UK is 11/1000 pregnancies and the mortality rate is around 10/100 000. Approximately 11 000 cases of ectopic pregnancies are diagnosed each year in the UK

Heterotopic pregnancy

is the simultaneous development of a pregnancy within and outside the uterine cavity.

Although the incidence of a heterotopic pregnancy in the general population is low (1:25 000–30 000), the incidence is significantly higher after in vitro fertilization (IVF) treatment (1 per cent).

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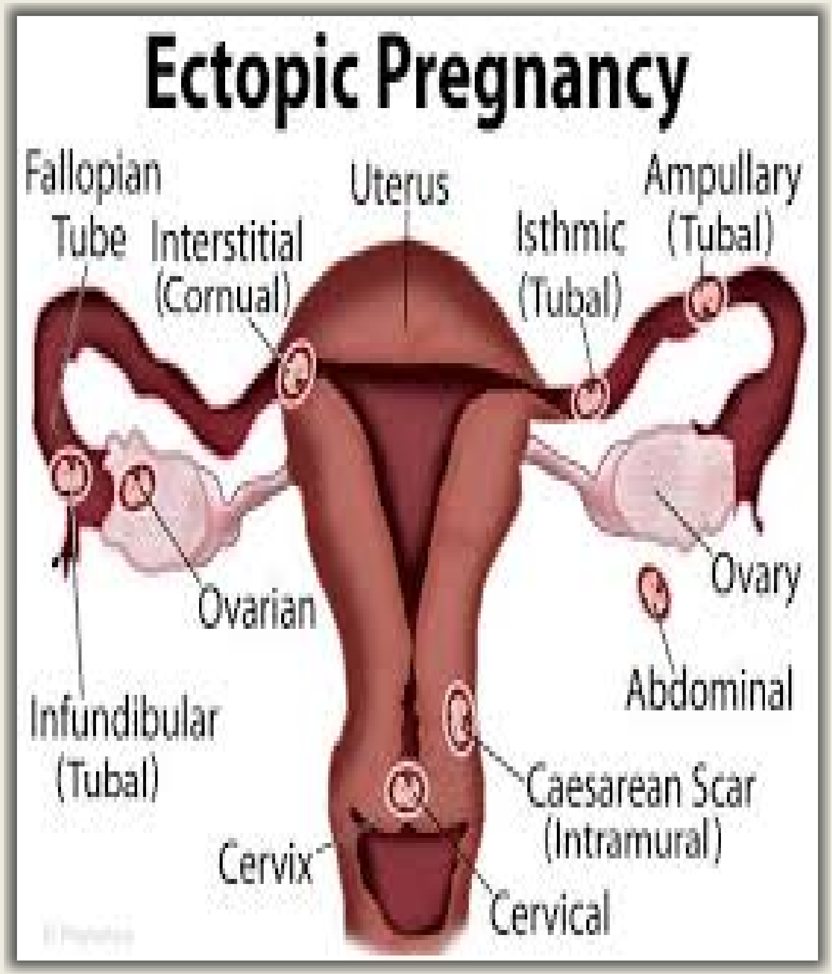
Site of ectopic

Common sites of implantation within the abdominal cavity are the Fallopian tubes (95 percent), ovaries (3 per cent) and peritoneal cavity (1 percent).

In the Fallopian tubes, the distribution of sites of implantation are the ampulla (74 per cent), isthmus (12 per cent), fimbrial end of the tube (12 per cent) and interstitium (2 per cent).



Known aetiological factors contributing to the risk of ectopic pregnancy are:

- tubal disease; pelvic infection, such as Chlamydia infection, has been estimated to account for 40 per cent of all ectopic pregnancies;
- previous ectopic pregnancy;
- previous tubal surgery;
- subfertility;
- use of intrauterine device.

Clinical presentation

1.Subacute :

The majority of patients present with a subacute clinical picture of abdominal pain and vaginal bleeding in early pregnancy. Vaginal bleeding is usually dark red, indicative of old blood. The

abdominal/pelvic pain may be localized to the iliac fossa. Occasionally, patients have shoulder tip pain indicative of free blood in the abdominal cavity causing diaphragmatic irritation and symptoms of dizziness (anaemia).

Bimanual examination can reveal tenderness in the fornices and there may be cervical excitation, again suggesting peritoneal irritation by free blood.

2. Acute presentation : some patients can present very acutely with rupture of the ectopic pregnancy and massive intraperitoneal bleeding. They can present with signs of hypovolaemic shock and acute abdomen.

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Investigation

The following are useful investigations for the diagnosis of ectopic pregnancy.

- Observations: BP, pulse, temperature
- Laboratory investigations: Hb, blood group and save (or crossmatch if patient is severely compromised) and BHCG.
- HCG: This hormone is a glycoprotein produced by the placenta. It has a half-life of up to 24 hours and peaks at around ten weeks. Pregnancy tests measure the B-subunit of HCG.

A bHCG level of less than 5 mIU/mL is considered negative for pregnancy, and anything above 25 mIU/mL is considered positive for pregnancy. In 85% of pregnancies, the bHCG levels almost double every 48 hours in a normally developing pregnancy. In patients with ectopic pregnancies, the rise of bHCG is often suboptimal. However, bHCG levels can vary widely in individuals and thus often multiple readings are required for comparison purposes.

- Transvaginal ultrasound scan (TVS):

An intrauterine gestational sac should be visualized at about 4.5 weeks of gestation. The corresponding bHCG at that gestation is around 1500 mIU/mL.

By the time a gestational sac with fetal heart pulsation is detected (at around 5 weeks gestation), bHCG level should be around 3000 mIU/mL.

Hence, the interpretation of bHCG must be done in context with the clinical picture and ultrasound findings.

Thus, if there were discrepancy between the bHCG concentrations and that seen on ultrasound scan (e.g. a high bHCG with no intrauterine pregnancy on ultrasound scan), the differential diagnosis of an ectopic pregnancy must be made.

Identification of an intrauterine pregnancy (gestation sac, yolk sac along with fetal pole) on TVS effectively excludes the possibility of an ectopic pregnancy in most patients except in those patients with rare heterotopic pregnancy.

The presence of free fluid during TVS is suggestive of a ruptured ectopic pregnancy.

- Laparoscopy: this can be used to diagnose and treat ectopic pregnancy.

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Management

Ectopic pregnancy can be managed using an expectant, medical or a surgical approach, depending on clinical presentation and patient choice.

Expectant

Expectant management is based on the assumption that a significant proportion of all tubal pregnancies will resolve through regression or a tubal abortion without any treatment.

This option is suitable for patients who are haemodynamically stable and asymptomatic. This entails serial bHCG measurements and ultrasonography

Medical

Systemic methotrexate is a treatment option for a carefully selected subgroup of patients. Methotrexate, a folic acid antagonist, inhibits DNA synthesis in trophoblastic cells.

The following are reasonable indications for the use of methotrexate:

(1) cornual pregnancy;

(2) persistent trophoblastic disease;

(3) patient with one Fallopian tube and fertility desired;

(4) patient who refuses surgery or in whom risks of surgery is too High;

(5) treatment of ectopic pregnancy where trophoblast is adherent to bowel or blood vessel.

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Protocol

1.Dx per guideline

2.Check criteria for medical Mx

3.Consent

4. Check Bl.uria, S.creatinine, LFT, Hb,Bl. group(**ANTI D**)

5.Calculate the dose (1mg/kg)

6.Follow up appointment.

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Single dose injection of methotrexate with success rate of about 94%

Tubal patency is found in 82% of cases after medical Rx

Dose $50mg/m^2$ B.wt, or 1mg/kg body wt.

Criteria for medical therapy

1.Haemodinamicly stable

2.No contraindication for meth.

3.Able to return for follow up

4.Adnexal mass < 5 cm

5.HCG level not exceed 5000(15000 under consultant agreement)

6.No fetal cardiac activity

7.Unfit for GA

8.No IU pregnancy

9.Non laparoscopic Dx of E.P

CONTRA INDICATIONS OF METHOTREXATE

1.Breast feeding

2.Immune deficiency/active infection

3.Active pul.,liver disease

4.Active peptic ulcer or colitis

5.Blood disorder

6.hepatic,renal or hemodynamic dysfunction

Side effects

1.nausea,vomiting,stomatits,

diarrhea,abd.discomfort

2.photosensitivity, skin reaction

3.Impaired liver function

4.Pneumonitis

5.Hematosalpinx

6.Sever pelvic pain between 5-7days

7.Reversible alopecia

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Treatment effect

1.Increase abdominal pain

2.Increase in HCG during 1st 3-days of treatment

3.Vaginal bleeding

Sign of treatment failure

1.Significantly worsening abd.pain

2.haemodinamic instability

3.HCG not decrease by 15% bet.4-7 days after treatment

4.Increasing or plateauing HCG level after 1st wk of Rx.

Counseling

1.Side effects& treatment effect discussed

2.report;- sever abd. Pain

-heavy vag.bleeding

- dizziness,syncope, palpitation

3.Avoid sunlight,alcohol,vitamines aspirin,NSAID, sex

Follow up

*Repeat HCG on day **5** post inj.

-if <**15**% decrease consider repeat dose &recheck in **5** days

-if >**15**% decrease, recheck weekly until HCG <25 iu

*lower abd.pain (separating pain)often worsens 2-6 days after injection, may occur.after several days but quickly resolve

*next period oc.befor HCG return to normal(contraception)

*4wk monitoring HCG till reach <5iu

Treatment failure

*its rare event mainly oc. when initial HCG >5000 iu

*use of Mifepristone 200mg,2nd dose of methotrexate with same follow up

If still not respond or fails then you have to do surgery

Surgical management

Surgical treatment can be by laparoscopy or laparotomy.

The laparoscopic approach offers significant advantage when compared to laparotomy as it results in less blood loss, shorter operating time, less analgesia requirement, a shorter hospital stay and a shorter convalescence than laparotomy. Laparoscopic surgery is the mainstay of management. Laparotomy is mainly reserved for severely compromised patients or due to the lack of endoscopic facilities.

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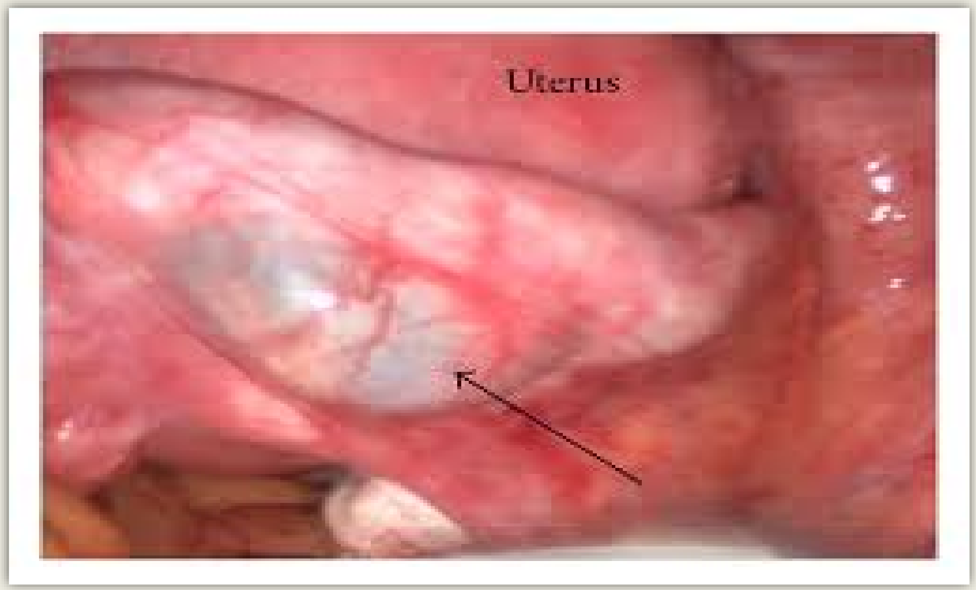
Surgical management

During surgery, the Fallopian tube can either be removed (salpingectomy)

or a small opening can be made at the site of the ectopic pregnancy and the trophoblastic tissue extracted via the opening (salpingotomy).

In general, if the patient has a normal remaining tube, salpingectomy is the treatment of choice.

Salpingotomy is thought to be associated with a higher rate of subsequent ectopic pregnancy.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